COMPETENCY BASED CURRICULUM

FOR

INFORMATION COMMUNICATION TECHNOLOGY

KNQF LEVEL 5

CYCLE 3

PROGRAMME ISCED CODE: 061 2454A

TVET CDACC

P.O. BOX 15745-00100

NAIROBI

Council Secretary/CEO
TVET Curriculum Development, Assessment and Certification Council
P.O. Box 15745–00100
Nairobi, Kenya
Email: info@tvetcdacc.go.ke

FOREWORD

The provision of quality education and training is fundamental to the Government's overall strategy for social and economic development. Quality education and training contribute to the achievement of Kenya's development blueprint and sustainable development goals.

Reforms in the education sector are necessary to achieve Kenya Vision 2030 and meet the provisions of the Constitution of Kenya 2010. The education sector had to be aligned to the Constitution, and this resulted in the formulation of the Policy Framework for Reforming Education and Training in Kenya (Sessional Paper No. 14 of 2012). A key feature of this policy is the radical change in the design and delivery of TVET training. This policy document requires that training in TVET be competency-based, curriculum development be industry-led, certification be based on demonstration of competence, and the mode of delivery allow for multiple entry and exit in TVET programmes.

These reforms demand that Industry takes a leading role in curriculum development to ensure the curriculum addresses its competence needs. It is against this background that this curriculum has been developed. For trainees to build their skills on foundational hands-on activities of the occupation, units of learning are grouped in modules. This has eliminated duplication of content and streamlined exemptions based on skills acquired as a trainee progresses in the up-skilling process, while at the same time allowing trainees to be employable in the shortest time possible through the acquisition of part qualifications.

It is my conviction that this curriculum will play a great role in developing competent human resources for the ICT Sector's growth and development.

PRINCIPAL SECRETARY

STATE DEPARTMENT FOR TVET

MINISTRY OF EDUCATION

PREFACE

Kenya Vision 2030 aims to transform Kenya into a newly industrializing middle-income country, providing high-quality life to all its citizens by the year 2030. Kenya intends to create globally competitive and adaptive human resource base to meet the requirements of a rapidly industrializing economy through lifelong education and training. TVET has a responsibility to facilitate the process of inculcating knowledge, skills, and worker behaviour necessary for catapulting the nation to a globally competitive country, hence the paradigm shift to embrace Competency-Based Education and Training (CBET).

TVET Act, CAP 210A and Sessional Paper No. 1 of 2019 on Reforming Education and Training in Kenya for Sustainable Development emphasized the need to reform curriculum development, assessment, and certification. This called for a shift to CBET to address the mismatch between skills acquired through training and skills needed by industry, as well as increase the global competitiveness of the Kenyan labour force.

This curriculum has been developed in adherence to the Kenya National Qualifications Framework and CBETA standards and guidelines. The curriculum is designed and organized into Units of Learning with Learning Outcomes, suggested delivery methods, learning resources, and methods of assessing the trainee's achievement. In addition, the units of learning have been grouped in modules to concretize the skills acquisition process and streamline upskilling.

I am grateful to all expert trainers and everyone who played a role in translating the Occupational Standards into this competency-based modular curriculum.

CHAIRPERSON, TVET CDACC

ACKNOWLEDGMENT

This curriculum has been designed for competency-based training and has independent units of learning that allow the trainee flexibility in entry and exit. In developing the curriculum, significant involvement and support were received from expert trainers, institutions and organizations.

I recognize with appreciation the role of the ICT National Sector Skills Committee (NSSC) in ensuring that competencies required by the industry are addressed in the curriculum. I also thank all stakeholders in the ICT sector for their valuable input and everyone who participated in developing this curriculum.

I am convinced that this curriculum will go a long way in ensuring that individuals aspiring to work in the ICT Sector acquire competencies to perform their work more efficiently and effectively.

COUNCIL SECRETARY/CEO

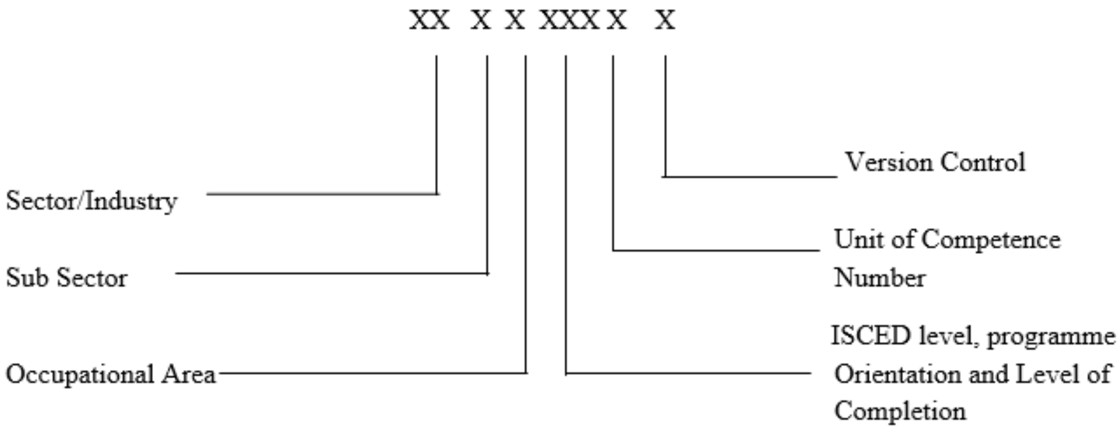
TVET CDACC

Table of Contents

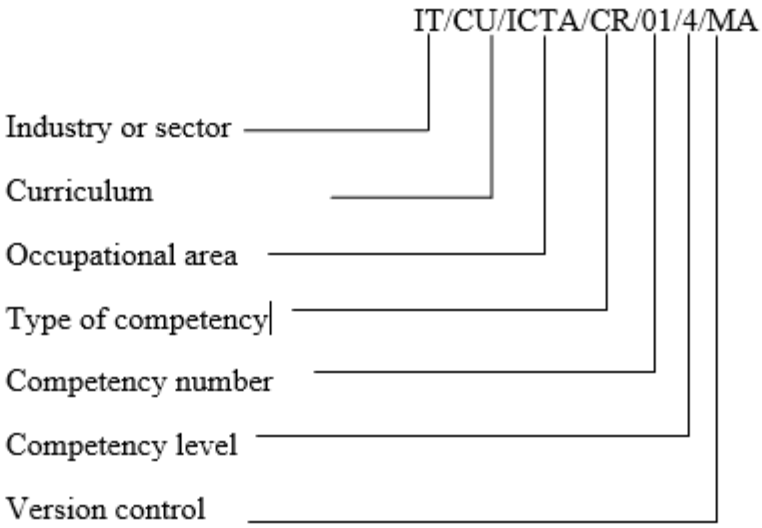
ACRONYMS

ICT	Information Communication Technology
IS	Information System
ISP	Information Security Policy
KCSE	Kenya Certificate of Secondary Education
KNQA	Kenya National Qualification Authority
KNQF	Kenya National Qualification Framework
LAN	Local Area Network
MIS	Management Information System
PAN	Personal Area Network
SOPs	Sum of Product
POST	Power on Self-Test
PPE	Personal Protective Equipment
RAM	Random Access Memory
SDLC	System Development life cycle
TVET	Technical and Vocational Education and Training
WAN	Wide Area Network
DOM	Document Object Model

KEY TO ISCED UNIT CODE



KEY TO TVET CDACC UNIT CODE



COURSE OVERVIEW

The ICT Technology Level 5 curriculum consists of competencies that an individual must have to supporting or enable the use of ICT equipment and applications.

It involves, performing computer essentials, performing computer operations, performing computer network setup and performing computer repair and maintenance, installing computer software, perform network design and management and managing computerized database system.

Summary of Units of Learning

ISCED Unit Code	TVET CDACC UNIT CODE	Units Title	Unit Duration (Hours)	Credit Factor
	MODULE I			
0611 351 01A	IT/CU/ICTA/CR/01/4/MA	Computer Essentials	120	12
0611 351 02A	IT/CU/ICTA/CR/02/4/MA	Computer Operations	150	15
Sub-Total Hours			**270**	**27**
MODULE II				
0612 351 03A	IT/CU/ICTA/CR/03/4/MA	Computer Network Setup	200	20
0714 351 04A	IT/CU/ICTA/CR/04/4/MA	Computer Repair and Maintenance	200	20
Sub-Total Hours			**400**	**40**
MODULE III				
0714 441 04A	IT/CU/ICTA/CC/01/5/MA	Basic Electronics	100	10
0619 451 06A	IT/CU/ICTA/CR/01/5/MA	Computer Software	160	16
0417 441 02A	IT/CU/ICTA/BC/01/5/MA	Work Ethics and Practice	40	4
0612 451 07A	IT/CU/ICTA/CR/02/5/MA	Network Design and Management	160	16
Sub-Total Hours			**460**	**46**

	MODULE IV			
0613 451 05A	IT/CU/ICTA/CC/02/5/MA	Computer Programming Principles	180	18
0612 451 08A	IT/CU/ICTA/CR/03/5/MA	Computerized Database System	200	20
0031 441 01A	IT/CU/ICTA/BC/01/5/MA	Communication Skills	40	4
0413 441 03A	IT/CU/ICTA/BC/02/5/MA	Entrepreneurial Skills	40	4
Sub Total			**460**	**46**
Industry Training			480	48
GRAND TOTAL			**2,070**	**207**

Entry Requirements

An individual entering this course should have any of the following minimum requirements:

a) Kenya Certificate of Secondary Education (KCSE) mean grade D(Plain) or completion of KNQF level 4

Or

b) Any other qualification equivalent to that of ICT Operator level 4 as determined by TVETA

Trainer Qualification

A trainer for any of the Units of Competency in this course must:

a) Have at least a minimum of ICT Technician KNQF Level 6 qualification or its equivalent in a trade area related to this course.
b) Be registered by TVETA.

Industry Training

An individual enrolled in this course will be required to undergo Industry training for a minimum period of 480 hours in ICT sector. The industrial training may be taken after completion of all units for those pursuing the full qualification or be distributed equally in each unit for that pursuing part qualification. In the case of dual training model, industrial training shall be as guided by the dual training policy.

Assessment

The course shall be assessed formatively and summative:

a) During formative assessment all performance criteria shall be assessed based on performance criteria weighting.
b) Summative assessment shall focus on critical aspects of the Unit of competency.
c) Theoretical and practical weighting for each unit of learning shall be as follows
    i. 10: 90 for unit in module one and module two
    ii. 30:70 for the units in module three and module four.
d) Formative and summative assessment weights shall constitute 60% and 40% of the overall score respectively.
e) For a candidate to be declared competent in a unit of competency, the candidate must meet the following conditions:
    i) Obtained at least 40% in theory assessment in formative and summative assessments.
    ii) Obtained at least 50% in practical assessment in formative and summative assessment where applicable.
    iii) Obtained at least 50% in the weighted results between formative assessment and summative assessment where the former constitutes 60% and the latter 40% of the overall score.
f) Assessment performance rating for each unit of competency shall be as follows:

MARKS	COMPETENCE RATING
80 -100	Attained Mastery
65 - 79	Proficient
50 - 64	Competent
49 and below	Not Yet Competent
Y	Assessment Malpractice/irregularities

g) Assessment for Recognition of Prior Learning (RPL) may lead to award of part and/or full qualification.

Certification

A candidate will be issued with a Certificate of Competency upon demonstration of competence in a core Unit of Competency. To be issued with the Kenya National TVET Certificate in Information Communication Technology level 5, the candidate must demonstrate competence in all the Units of Competency as given in the qualification pack. Statement of Attainment certificate may be awarded upon demonstration of competence in certifiable element within a unit.

These certificates will be issued by TVET CDACC

MODULE 1

UNIT CATEGORY	ISCED UNIT CODE	TVET CDACC UNIT CODE	UNITS NAME	DURATION (HOURS)
CORE	0611 351 01A	IT/CU/ICTA/CR/01/4/MA	Computer Essentials	120
CORE	0611 351 02A	IT/CU/ICTA/CR/02/4/MA	Computer Operations	150
Total hours				**270**

COMPUTER ESSENTIALS

ISCED UNIT CODE: 0611 351 01A

TVET CDACC UNIT CODE: IT/CU/ICTA/CR/01/4/MA

Duration of unit: 120 hours

Relationship to Occupational Standards

This unit addresses the unit of competency: Perform Computer Essentials

Unit Description

This unit covers the competencies required in performing computer essentials. It involves managing computer devices, managing desktop settings, performing file management, managing computer software and performing online jobs.

Summary of Learning Outcomes

Learning Outcomes	Durations(Hours)
1. Manage computer devices	20
2. Manage desktop settings	30
3. Perform file management	20
4. Manage computer software	20
5. To Perform online jobs	30
Total Hours	**120**

Learning outcomes, Content and Suggested Assessment Methods

Learning outcome	Content	Suggested Assessment Methods
1. Manage computer devices	1.1. Selection of Computer Hardware devices 1.1.1. Introduction to computer devices 1.1.1.1. Meaning of computer hardware devices	• Practical • Oral questions • Written tests

	1.1.1.2. Identification of computer components and port 1.1.2. Computer case, monitor, keyboard, and mouse 1.1.3. All the parts inside the computer case, such as the hard disk drive, motherboard and video card s 1.1.3.1. Classification of computer hardware devices 1.2. Disassembling of computer hardware devices 1.2.1. Cleaning of computer devices 1.3. Assembling of Computer Hardware devices 1.3.1. Types of Computer Hardware devices 1.3.2. Functions of various computer hardware devices 1.3.3. Connecting computer hardware devices e.g. monitor, System Unit 1.4. Booting of computer 1.4.1. Introduction to booting 1.4.2. Types of booting 1.4.2.1. Cold Booting 1.4.2.2. Warm booting 1.5. Connecting computer peripheral devices 1.5.1. Types of computer peripheral devices 1.1.1.1. Printer 1.1.1.2. Speaker 1.1.1.3. Mouse 1.1.1.4. Keyboard 1.1.1.5. Projector 1.5.2. Configuration of peripheral devices	• Observation • Reports • Portfolio of evidence
2. Manage	2.1 Customization of desktop icons	• Practical

desktop settings	2.1.1 Introduction to desktop icons and settings 2.2 Date and time settings 2.3 Desktop settings customization 2.3.1 Background colour and pictures 2.3.2 Themes 2.3.3 Taskbar 2.3.4 Menu bar 2.3.5 Text size 2.3.6 Brightness	• Oral questions • Written tests • Observation • Reports • Portfolio of evidence
3. Perform file management	3.1 Creating files and folders 3.1.1 Introduction to computer files and folders 3.1.2 Creation of files and folders 3.1.3 Compression and extraction of folders 3.2 Transferring files and folders 3.2.1 sharing of folders and files 3.3 File protection 3.3.1 Password 3.3.2 Encryption	• Practical • Oral questions • Written tests • Observation • Reports • Portfolio of evidence
4. Manage computer software	4.1 Selecting data backup media 4.1.1 Types of data Backup media 4.2 Performing data backup 4.3 Installation of computer software 4.3.1 Introduction to computer software 4.3.2 Types of computer software 4.3.2.1 Applications 4.3.2.2 Operating systems 4.3.2.3 Utility programs 4.3.3 Configuration of computer software 4.4 Optimization of computer software 4.4.1 Updating computer software 4.4.2 Computer disk cleanup	• Practical • Oral questions • Written tests • Observation • Reports • Portfolio of evidence

5. Perform Online Jobs	5.1. Introduction to online working 5.1.1. Types of online Jobs 5.1.2. Online job platforms (Upwork, Freelancer, Fiverr) 5.2. Online account and profile management 5.3. Identifying online jobs job bidding 5.4. Online digital identity 5.5. Online job bidding 5.6. Executing online tasks 5.7. Management of online payment accounts.	• Practical Assessment • Project • Third Party Report • Portfolio of Evidence • Written Assessment • Oral Questioning

Suggested Delivery Methods

- Demonstration by trainer
- Practical work by trainee
- Viewing of related videos
- Group discussions
- Direct instructions

Recommended resources for 25 trainees

S/No.	**Category/Item**	**Description/ Specifications**	**Quantity**	**Recommended Ratio** (Trainee: Item)
A	**Learning Materials**			
1.	Textbooks		5 pcs	5:1
2.	Installation manuals		5 pcs	5:1
3.	Flip Charts		5 pcs	5:1
4.	PowerPoint presentations	For trainer’s use		

5.	Installation CDs/DVDs			
B	**Learning Facilities & infrastructure**			
6.	Lecture/theory room		1	25:1
7.	Computer laboratory		1	25:1
C	**Consumable materials**			
8.	Printing papers		1 ream	1:20
9.	Foolscaps		1 ream	1:20
10.	Toners		2 pcs	13:1
11.	Assorted colour of whiteboard markers			
D	**Tools and Equipment**			
12.	Computers		25 pcs	1:1
13.	Projector		1 pcs	25:1
14.	Printers		2 pcs	13:1
15.	Whiteboard		1 pcs	25:1
16.	Flash drives		5 pcs	5:1
17.	External Hard drive		5 pcs	5:1
18.	System Software suite		5 pcs	5:1
19.	Application Software suite		5 pcs	5:1
20.	Computer Repair Tool box		5	5:1

COMPUTER OPERATIONS

ISCED UNIT CODE: 0611 351 02A

TVET CDACC UNIT CODE: IT/CU/ICTA/CR/02/4/MA

Duration of Unit: 150 hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform Computer Operations

Unit Description

This unit covers the competencies required to perform computer operations. It involves processing computerized word documents, manipulating computerized spreadsheets, maintaining computerized databases, preparing PowerPoint presentation slides, manipulating graphic application and performing online collaboration.

Summary of Learning Outcomes

Learning Outcomes	Durations(Hours)
1. Process computerized word document	30
2. Manipulate computerized spreadsheet	30
3. Maintain computerized database	30
4. Prepare PowerPoint presentation	20
5. Manipulate graphic application	25
6. Perform online collaboration	15
Total Hours:	**150**

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Process computerized word	1.1 Ergonomics risk factors 1.2 Creation of computerized word	● Practical assessment ● Simulations

document	document 1.2.1 Introduction to word document 1.2.2 Types of word processors 1.2.3 Creating word document 1.2.4 Editing and formatting word document 1.2.5 Word document editing features 1.2.5.1 Text editing 1.2.5.2 Paragraph editing 1.2.5.3 Document editing 1.2.6 Word document formatting features 1.2.6.1 Text formatting 1.2.6.2 Paragraph formatting 1.2.6.3 Document formatting 1.2.7 Enhancing productivity 1.2.7.1 Set basic options/ preferences 1.2.7.2 Help resources 1.2.7.3 Use magnification/zoom tools 1.2.7.4 Display, hide built-in tool bar 1.3 Creation and manipulation of tables 1.3.1 Inserting tables 1.3.2 Working with tables 1.4 Mail merge 1.5.1 Mail merge preparation 1.5.2 Mail merge output 1.5 Inserting word processing objects	• Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence

	1.5.1 Picture 1.5.2 Shapes 1.5.3 Table 1.5.4 Charts 1.6 Generating list of figures and table of content 1.6.1 List of figures 1.6.2 Table of content 1.7 Printing of computerized word document 1.7.1 Print setup 1.7.2 Printing	
2. Manipulate computerized spreadsheet	2.1 Creation of Computerized spreadsheet workbook 2.1.1 Spreadsheet concepts 2.1.2 Elements of spreadsheet window 2.1.2.1 Worksheet 2.1.2.2 workbook 2.1.2.3 Rows 2.1.2.4 columns 2.1.2.5 Cells 2.2 Cell referencing 2.2.1.1 Relative cell referencing 2.2.1.2 Absolute cell referencing 2.2.1.3 Mixed cell referencing 2.2.2 Spreadsheet editing features	• Practical assessment • Simulations • Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence

	2.2.2.1 Worksheet editing 2.2.2.2 Inserting rows/columns 2.2.2.3 Removing rows/columns 2.2.2.4 Adjusting row heights and column width 2.2.2.5 Inserting worksheets 2.2.2.6 Renaming worksheets 2.2.2.7 Move or copy worksheets 2.2.2.8 Deleting worksheets 2.2.3 Data manipulation in spreadsheets 2.2.3.1 Data entry 2.2.3.2 Types of data 2.3 Formulas and functions 2.3.1.1 Formulas and functions syntax 2.3.1.2 Arithmetic functions 2.3.1.3 logical functions 2.3.1.4 Look up functions 2.3.2 Computerized spreadsheet worksheet formatting 2.3.2.1 Font styles	

	2.3.2.2 Alignment 2.3.2.3 Borders and shading 2.3.2.4 Header and footer 2.4 Charts generation 2.4.1.1 Types of charts 2.4.1.2 Insert charts 2.4.1.3 Labelling and Editing charts 2.4.1.4 Computerized spreadsheet workbook printing 2.4.1.5 Print setup 2.4.1.6 Printing	
3. Maintain computerised database	3.1 Computerised database user requirements collection 3.1.1 Introduction to database 3.1.1.1 Key concepts 3.1.1.2 Database organisation 3.1.1.3 Database relationships 3.1.1.4 Database operations 3.1.2 Collection of User requirements 3.2 Design Computerised database schema 3.2.1 Creating database models 3.2.1.1 ERD models 3.2.1.2 Relational models	• Practical assessment • Simulations • Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence

	3.3 Creation of Computerised database objects 3.3.1 Database Objects 3.3.1.1 Tables 3.3.1.2 Records 3.3.1.3 Fields 3.3.1.4 Keys 3.3.1.5 Forms 3.3.1.6 Queries 3.3.1.7 Reports 3.4 Data manipulation 3.4.1 Inserting records 3.4.2 Retrieving records 3.4.3 Deleting records 3.4.4 Updating record 3.4.5 Printing database objects 3.4.5.1 Tables 3.4.5.2 Forms 3.4.5.3 Queries 3.4.5.4 Reports	
4. Prepare Power point presentation	4.1 Collecting PowerPoint Presentation requirements 4.1.1 Definition of terms 4.1.2 Presentation requirements 4.1.3 Types of presentation software 4.1.4 Elements of presentation window 4.2 Creating PowerPoint slides 4.2.1 Types of presentation layout	• Practical assessment • Simulations • Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence

	4.2.2 Factors to consider when designing presentation layout 4.2.3 Design a PowerPoint presentation 4.2.4 Create a PowerPoint presentation 4.2.5 Save a PowerPoint presentation 4.3 Exhibit presentation views 4.2.1 Slide views 4.2.2 Working with presentations 4.3.1.1 Switch between open PowerPoint presentations 4.4 Perform animation and transitions 4.4.1 Slide animation 4.4.2 Slide transition 4.5 Manipulation of PowerPoint slides 4.5.1 Adding data/text to a slide 4.5.2 Formatting data/text 4.5.3 Move/copy/delete a slide 4.5.4 Inserting header and footer 4.5.5 Presentation objects 4.5.5.1 Tables 4.5.5.2 Charts 4.6 Printing of PowerPoint slides 4.6.1 Print setup 4.6.2 Printing PowerPoint	

	presentation	
5. Manipulate graphic application	5.1 Identifying graphic design requirements 5.1.1 Definition of terms 5.1.2 Graphic application requirements 5.1.3 Types of graphic application software 5.1.4 Types of publications designs 5.1.4.1 Templates 5.1.4.2 Banners 5.1.4.3 Booklets 5.1.4.4 Brochures 5.1.4.5 Flyers 5.1.4.6 Posters 5.1.4.7 Cards 5.1.4.8 Certificates 5.1.4.9 Magazines 5.1.5 Elements of Graphic application window 5.2 Creation of graphic design 5.2.1 Perform basic tasks using graphic application software 5.2.1.1 Publication type 5.2.1.2 Page setup 5.2.1.3 Ruler/guides 5.2.1.4 Page views 5.2.2 Add content to a	• Practical assessment • Simulations • Project • Written assessment • Portfolio of evidence

	publication 5.2.3 Edit content to a publication 5.2.4 Format text and paragraphs in a publication 5.2.5 Page formatting in a publication 5.2.5.1 Columns 5.2.5.2 Borders and shading 5.2.5.3 Headers and footers 5.2.5.4 Background 5.2.5.5 Watermarks 5.2.5.6 Orientation 5.2.6 Work with graphics objects in a publication 5.2.6.1 Textbox 5.2.6.2 Tables 5.2.6.3 Shapes 5.2.6.4 Pictures 5.2.6.5 (PNG, JPEG, GIF) 5.3 Publishing of graphic design 5.3.1 Prepare a publication 5.3.2 Print setup 5.3.3 Printing publication	
6. Perform document production	6.1 Printing documents 6.1.1 Introduction to document production 6.1.2 Types of printers 6.1.3 Document printing	• Practical assessment • Simulations • Project • Observation Checklist • Product Checklist

	6.2 Document scanning 6.2.1 Types of scanners 6.2.2 Document scanning 6.3 Document duplication	• Written assessment • Portfolio of evidence
7. Perform Online Collaboration	7.1 Identification of Online collaboration tools 7.1.1 Definition of online collaboration 7.1.2 Importance of online collaboration 7.1.3 Factors to consider when choosing an online collaboration tool 7.1.4 Online collaboration tools 7.1.4.1 Microsoft teams 7.1.4.2 Skype 7.1.4.3 Google drive 7.1.4.4 Zoom 7.1.4.5 Google meet 7.1.4.6 Slack 7.2 Online collaboration preparation 7.2.1 Online collaboration key concepts 7.2.2 Common setup features 7.2.2.1 Download software to support online collaboration tools 7.2.2.2 Register and/ or set a user account 7.2.3 Preparation for online collaboration	• Practical assessment • Simulations • Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence

	7.3 Application of online collaborative tools 7.3.1 Using online collaborative tools 7.3.1.1 Online storage media 7.3.1.2 Using email 7.3.1.2.1 Sending and receiving email 7.3.1.2.2 Tools and settings 7.3.1.2.3 Organizing email 7.3.1.3 Using calendars 7.3.1.4 Online calendars 7.3.1.5 Social media 7.3.1.6 Online learning environment 7.3.1.7 Synchronization tools 7.4 Demonstrating Mobile collaborations 7.4.1 Key concepts in mobile applications 7.4.2 Mobile applications permissions 7.4.3 Synchronization	

Suggested Delivery Methods

- Demonstration by trainer
- Practical work by trainee
- Viewing of related videos
- Group discussions
- Facilitation using active learning strategies

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/	Quantity	Recommended

		Specifications		Ratio (Trainee: Item)
A	**Learning Materials**			
1.	Textbooks		5 pcs	5:1
2.	Installation manuals		5 pcs	5:1
3.	Flip Charts		5 pcs	5:1
4.	PowerPoint presentations	For trainer's use		
5.	Magazines/brochures/business cards			
B	**Learning Facilities & infrastructure**			
6.	Lecture/theory room		1	25:1
7.	Laboratory		1	25:1
C	**Consumable materials**			
8.	Printing papers		1 ream	1:20
9.	Foolscaps		1 ream	
10.	Toners/cartridges		2 pcs	13:1
11.	Assorted colour of whiteboard markers			
D	**Tools and Equipment**			
12.	Computers		25 pcs	1:1
13.	Projector		1 pc	25:1

14.	Printers		2 pcs	1:13
15.	Whiteboard		1 pc	25:1
16.	Flash drives		5 pcs	5:1
17.	1 External Hard drive		1 pcs	25:1
18.	Application software suite		5 pcs	5:1

MODULE 2

UNIT CATEGORY	ISCED UNIT CODE	TVET CDACC UNIT CODE	UNITS NAME	DURATION (HOURS)
CORE	0612 351 03A	IT/CU/ICTA/CR/03/4/MA	Computer Network Setup	200
CORE	0714 351 04A	IT/CU/ICTA/CR/04/4/MA	Computer Repair and Maintenance	200
	Sub-Total			**400**
	Industrial Training			**320**
	Total Hours			**720**

COMPUTER NETWORK SETUP

ISCED UNIT CODE: 0612 351 03A

TVET CDACC UNIT CODE: IT/CU/ICTA/CR/03/4/MA

Duration of unit: 200 hours

Relationship to Occupational Standards

This unit addresses the unit of competency: Setup Computer Network

Unit Description:

This unit covers the competencies required in setup computer network. It involves the ability to terminate network cables, connect network cables and perform computer network Maintenance.

Summary of Learning Outcomes

Learning Outcomes	Duration (Hours)
1. Terminate Computer network cables	70
2. Connect Computer network cables	70
3. Perform Computer network Maintenance	60
Total Hours	**200**

Learning outcomes, Content and Suggested Assessment Methods

Learning outcome	Content	Suggested Assessment Methods
1. Terminate Computer network cables	1.1 Selecting Network devices 1.1.1 Introduction to computer networks 1.1.2 Types of network topologies	• Practical • Oral questions • Written tests • Observation

	1.1.3 Types of network devices 1.1.4 Components of a computer networks 1.1.5 Types of network tools 1.1.6 Cable colour coding 1.2 Network cable trunking 1.2.1 Definition cable trunking 1.2.2 Types of cable trunking 1.2.3 Tools used in cabling trunking 1.2.3.1 Measuring tape 1.2.3.2 Pencil 1.2.3.3 Cable ties 1.2.3.4 Wire cutters 1.2.3.5 Safety equipment 1.2.3.6 Spirit level 1.2.3.7 Drill 1.2.3.8 Screwdriver 1.3 Network cable termination 1.3.1 Definition of networking cable termination 1.3.2 Tools for cable termination 1.3.2.1 RJ45 connectors 1.3.2.2 Crimping tool 1.3.2.3 Wire stripper 1.3.2.4 Cable cutter 1.3.2 Process of cable termination 1.3.2.1 Cable stripping 1.3.2.2 Colour coding 1.3.2.3 Cable crimping	• Portfolio of evidence
2. Connect Computer	2.1 Observing safety measures in networking	• Practical • Oral questions

network cables	2.1.1 Computer network safety measures 2.1.1.1 Overall/apron/dust coat 2.1.1.2 Gloves 2.1.1.3 Safety boots 2.1.1.4 Ergonomics 2.1.1.5 First AID kit 2.2 Setup network devices 2.4.1 Router 2.4.2 Switch 2.4.3 Bridge 2.4.4 Hub 2.4.5 Patch panels 2.4.6 Access point 2.3 Network cable testing 2.3.1 Cable testing methods 2.3.2 Continuity Testing 2.3.3 Wire Mapping 2.3.4 Cable Length Testing 2.3.5 Fault Detection 2.3.6 Cable testing tools 2.3.6.1 Cable tester 2.3.6.2 Multimeter 2.3.6.3 Crimping tool 2.3.6.4 Wire Stripper and cutter 2.4 Network cable connection 2.4.1 Networking standards 2.4.1.1 HTTP 2.4.1.2 IEEE 802.1 2.4.1.3 TCP/IP 2.5 Network connection establishment	• Written tests • Observation • Portfolio of evidence

	2.6 Network testing	
3. Perform Computer Network Maintenance	3.1 Monitoring computer network 3.1.1 Introduction to computer network monitoring and maintenance 3.1.2 Computer network monitoring physical tools 3.1.2.1 Cable testers 3.1.2.2 Crimping tool 3.1.2.3 Stripping tool 3.1.3 Physical networking device status monitoring 3.1.3.1 Port and interface 3.1.3.2 Cable and connection 3.1.3.3 Power supply 3.1.3.4 Network optimization 3.2 Troubleshooting Computer network 3.3 Optimizing Computer network 3.3.1 Upgrading network hardware devices 3.3.2 Upgrading computer network cables	• Practical • Oral questions • Written tests • Observation • Portfolio of evidence

Suggested Delivery Methods

- In Instructor led facilitation of theory
- Demonstration by trainer
- Practical work by trainee
- Viewing of related videos
- Group discussions

- Simulation

Recommended resources for 25 trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Trainee: Item)
A	Learning Materials			
1.	Textbooks		13 pcs	13:1
2.	Installation manuals		5pcs	5:1
3.	Charts			
4.	PowerPoint presentations	For trainer's use		
B	Learning Facilities & infrastructure			
5.	Lecture/theory room		1	25:1
6.	Computer Laboratory		1	25:1
7.	Internet Connection			
C	Consumable materials			
8.	Printing papers		1 ream	1:20
9.	Toners		2 pcs	13:1
10.	Assorted colour of whiteboard markers			
D	Tools and Equipment			
1.	Computers		25 pcs	1:1
2.	Projector		1 pc	25:1
3.	Signal testers		5 pcs	5:1

4.	Header checker		25 pcs	1:1
5.	Crimping tools		25 pcs	1:1
6.	Cable tester		5 pcs	5:1
7.	Switches		5pcs	5:1
8.	Repeaters		5pcs	5:1
9.	Routers/modem		5pcs	5:1
10.	Network tool kit		25 pcs	1:1
11.	RJ45		300 pcs	1:10
12.	UTP Ethernet Cable		300 metres	1:10
13.	Antistatic gloves		25 pairs	1:1

COMPUTER REPAIR AND MAINTENANCE

ISCED UNIT CODE: 0714 351 04A

TVET CDACC UNIT CODE: IT/CU/ICTA/CR/04/4/MA

Duration of Unit: 200 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform Computer Repair and Maintenance

Unit Description

This unit covers the competencies required for performing computer repair and maintenance. It involves performing computer troubleshooting, repairing faulty components, testing computer component functionality and performing computer maintenance.

Summary of Learning Outcomes

Learning Outcomes	Durations (Hours)
1. Perform computer troubleshooting	50
2. Repair faulty components.	60
3. Test computer component functionality	60
4. Perform computer maintenance	30
Total Hours	**200**

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Perform computer troubleshooting	1.1. User data assessment 1.1.1. Introduction to computer repair and maintenance 1.1.2. Documenting faulty computer user	• Practical assessment • Project • Observation

	data 1.2. Computer problems identification 1.2.1. Computer troubleshooting approaches 1.2.2. Basic computer hardware faults 1.2.3. Methods of information gathering 1.2.4. User data analysis 1.3. Determining solution to the problem 1.3.1. Computer hardware faults remedies 1.3.2. Test hypothesis 1.3.3. Problem Identification 1.3.4. Documentation of solution	Checklist • Product Checklist • Written assessment • Portfolio of evidence
2. Repair faulty components.	2.1 Selection of computer components for replacement 2.1.1 Computer hardware components 2.1.1.1 Factors to consider in selecting computer components 2.1.1.2 computer hardware components parts acquisition 2.2 Assembly of tools for repairing or replacing 2.2.1 Computer repair and maintenance tools 2.2.1.1 Straight-head screwdriver, large and small 2.2.1.2 Phillips-head screwdriver, large and small 2.2.1.3 Tweezers or part retriever 2.2.1.4 Needle-nosed pliers 2.2.1.5 Wire cutters 2.2.1.6 Chip extractor	• Practical assessment • Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence

	2.2.1.7 Hex wrench set 2.2.1.8 Torx screwdriver 2.3 Observation of Safety procedures 2.3.1 Safety measures and procedures 2.3.1.1 Personal Protective Equipment’s 2.3.1.2 Proper use of tools and equipment 2.3.1.3 Fire safety 2.3.1.4 Classes of fires 2.3.1.5 Fire extinguishers 2.3.1.6 Emergency procedures 2.3.1.7 First AID kit 2.3.1.8 Emergency contact 2.3.1.9 Contingency measures 2.4 Repair and replacing computer components 2.4.1 Computer components Instruction manuals 2.4.2 Computer components disassembly process 2.4.3 Reassembling repaired or replaced computer components 2.5 Disposing faulty or obsolete computer hardware components 2.5.1 Pollution 2.5.2 E- waste 2.5.3 Hazards 2.5.4 Types of E-waste 2.5.5 Proper disposal methods	
3. Test computer component functionality	3.1 Performing POST on computer 3.2 Performing computer component test 3.2.1 Importance of testing 3.2.2 Testing techniques	• Practical assessment • Project • Observation

	3.2.2.1 Testing of repaired or replaced components 3.2.3 Evaluation of test Results 3.3 Computer component's functionality report 3.3.1 Generation of test results report	Checklist • Product Checklist • Written assessment • Portfolio of evidence
4. Perform computer maintenance	4.1 Computer maintenance scheduling 4.1.1 Introduction to computer maintenance 4.1.1.1 Definition of computer maintenance 4.1.1.2 Importance of computer maintenance 4.1.2 Types of computer maintenance 4.1.3 Prepare computer maintenance schedule 4.2 Performing computer maintenance 4.2.1 Computer maintenance utilities 4.2.2 Uses of computer maintenance utilities 4.2.3 Perform computer maintenance 4.3 Computer maintenance report 4.3.1 Importance of computer maintenance report 4.3.2 Components of computer maintenance report	• Practical assessment • Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence

Suggested Delivery Methods

- Instructor led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainee

- Viewing of related videos
- Group discussions
- Direct instructions

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Trainee: Item)
A	Learning Materials			
1.	Textbooks		5 pcs	5:1
2.	Installation manuals		5 pcs	5:1
3.	Flip Charts		5 pcs	5:1
4.	PowerPoint presentations	For trainer's use		
5.	Magazines/brochures/business cards			
B	Learning Facilities & infrastructure			
6.	Lecture/theory room		1	25:1
7.	Computer Laboratory		1	25:1
C	Consumable materials			
8.	Printing papers		1 ream	1:20
9.	Foolscaps		1 ream	
10.	Toners		2 pcs	13:1
11.	Assorted colour of			

	whiteboard markers			
D	Tools and Equipment			
12.	Computers		25 pcs	1:1
13.	Projector		1 pcs	25:1
14.	Printers		2 pcs	13:1
15.	Whiteboard		1 pcs	25:1
16.	Flash drives		5 pcs	5:1
17.	1 External Hard drive		1 pcs	25:1
18.	Computer Repair Tool box		5	5:1

MODULE 3

UNIT CATEGORY	ISCED UNIT CODE	TVET CDACC UNIT CODE	UNIT NAME	DURATION (HOURS)
COMMON	0714 441 04A	IT/CU/ICTA/CC/01/5/MA	Basic Electronics	100
CORE	0619 451 06A	IT/CU/ICTA/CR/01/5/MA	Computer Software	160
BASIC	0417 441 02A	IT/CU/ICTA/BC/01/5/MA	Work Ethics and Practice	40
CORE	0612 451 07A	IT/CU/ICTA/CR/02/5/MA	Network Design and Management	160
Total Hours				**460**

BASIC ELECTRONICS

ISCED UNIT CODE: 0714 441 04A

TVET CDACC UNIT CODE: IT/CU/ICTA/CC/01/5/MA

Duration of Unit: 100 Hours

Relationship to Occupational Standards

This unit addresses the unit of competency: Apply Basic Electronics

Unit description

This unit specifies the competencies required to apply basic electronic. It involves identifying electric circuits, identifying electronic components, applying semi-conductor theory, and classifying computer memory, applying logic gates, applying logic gates and performing Boolean algebra operations.

Summary of Learning Outcomes

Learning Outcomes	Duration (hours)
1. Identify electric circuits	10
2. Identify electronic components	10
3. Apply semi-conductor theory	20
4. Classify computer memory	10
5. Apply logic gates	30
6. Perform Boolean algebra operations	20
Total Hours	**100**

Learning Outcomes, Content, and Suggested Assessment Methods

Learning outcomes	Content	Suggested Assessment

		Methods
1. Identify electrical circuits	1.1 Electrical circuit identification 1.1.1 Definition of electrical circuit 1.1.2 Components of electrical circuit 1.2 Electrical quantities and their S.I units' identification 1.2.1 Basic electrical quantities and their units 1.2.1.1 **Emf** in volts 1.2.1.2 Current in Amperes 1.2.1.3 Power in watts 1.2.1.4 Energy in joules 1.2.1.5 Resistance in ohms 1.3 Types of electrical circuits 1.3.1 AC – Alternating Current 1.3.2 DC – Direct Current	• Practical Activities • Project work • Demonstration • Group discussions • Observation • Third Party report • Portfolio of Evidence • Written tests
2. Identify Electronic components	2.1 Identification of electronic components 2.1.1 Resistor 2.1.2 Capacitor 2.1.3 Diode 2.1.4 Inductor 2.2 Characteristic of electronic components. 2.3 Application of electronic components. 2.4 Characteristics of integrated circuit	• Practical Activities • Project work • Demonstration • Group discussions • Observation • Third Party report • Portfolio of Evidence • Written tests
3. Apply semi-conductor	3.1 Explanation of semiconductor theory 3.2 Descriptions of structure of matter	• Practical Activities

theory	3.3 Explanation of Electrons in conductors and semiconductors 3.4 Types of semiconductor materials 3.4.1 Silicon 3.4.2 germanium 3.5 Explanation of P-type and N-type materials 3.6 Description of P-N junction diodes 3.6.1 Forward biasing 3.6.2 Reverse biasing 3.7 Types and operations of transistors 3.7.1 PNP type 3.7.2 NPN type 3.8 Application of Semiconductor theory	• Project work • Demonstration • Group discussions • Observation • Third Party report • Portfolio of Evidence • Written tests
4. Classify computer memory	4.1 Identification of computer memories 4.1.1 Definition of computer memory 4.1.2 Classification of computer memory 4.1.2.1 Primary memory 4.1.2.2 Secondary memory 4.1.3 Types of computer memories 4.1.3.1 RAM 4.1.3.2 ROM 4.1.3.3 DAM 4.2 Identification of Memory hierarchy speed 4.2.1 Registers 4.2.2 Cache memory 4.2.3 Main memory 4.2.4 Secondary storage 4.2.5 Tertiary storage 4.3 Identification of memory storage levels	• Practical Activities • Project work • Demonstration • Group discussions • Observation • Third Party report • Portfolio of Evidence • Written tests

	4.3.1 Internal 4.3.2 Main 4.3.3 Online 4.3.4 Offline bulk 4.4 Classify computer memories as per the technology used 4.4.1 Semiconductor memory 4.4.2 Magnetic memory 4.4.3 Optical memory	
5 Apply logic gates	5.1 Identification of Logic gates 5.1.1 Definition of terms 5.1.2 Types of logic gates 5.1.2.1 AND Gate 5.1.2.2 OR Gate 5.1.2.3 NOT Gate 5.1.2.4 NAND Gate 5.1.2.5 NOR Gate 5.1.2.6 XOR Gate 5.1.2.7 XNOR Gate 5.2 Development of Logic circuits 5.3 Simplification of Logic circuits 5.3.1 Logic circuits Simplification Methods 5.3.1.1 Boolean Algebra 5.3.1.2 K-Maps 5.3.1.3 Quine-McCluskey Algorithm 5.3.1.4 Software and CAD Tools 5.4 Application of logic gates in electronic circuits	• Practical Activities • Project work • Demonstration • Group discussions • Observation • Third Party report • Portfolio of Evidence • Written tests
6 Perform Boolean algebra	6.1 Key concepts in Boolean algebra 6.1.1 Boolean variables 6.1.2 Logical operations	• Practical Activities • Project work

operations	6.1.3 Boolean expressions 6.1.4 Laws and rules of Boolean algebra 6.1.5 Truth tables 6.1.6 De Morgan's theorem 6.2 Demonstration of Boolean expressions as per the SOPs 6.3 Performance of Basic Boolean operations 6.4 Methods of simplifying Boolean expressions 6.5 Illustration of Boolean Laws and Theorems 6.6 Simplification rules for Boolean expressions	• Demonstration • Group discussions • Observation • Third Party report • Portfolio of Evidence • Written tests

Suggested Delivery Methods

- Instructor led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainee
- Viewing of related videos
- Group discussions
- Direct instructions

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Trainee: Item)
A	Learning Materials			
1.	Textbooks		5 pcs	5:1
2.	Installation manuals		5 pcs	5:1
3.	Flip Charts		5 pcs	5:1
4.	PowerPoint presentations	For trainer's use		

5.	Magazines/brochures/business cards			
B	Learning Facilities & infrastructure			
6.	Lecture/theory room		1	25:1
7.	Laboratory		1	25:1
C	Consumable materials			
8.	Printing papers		1 ream	1:20
9.	Foolscaps		1 ream	
10.	Toners		2 pcs	13:1
11.	Assorted colour of whiteboard markers			
D	Tools and Equipment			
12.	Computers		25 pcs	1:1
13.	Projector		1 pcs	25:1
14.	Printers		2 pcs	13:1
15.	Whiteboard		1 pcs	25:1
16.	Ohmmeter		5	5:1
17.	Ammeter		5	5:1
18.	Digital Multi meter		5	5:1
19.	Power supplies		5	5:1
20.	Circuits		5	5:1

21.	Semiconductor materials		10	3:1
22.	Conductors e.g., copper, gold, silver		25	1:1
23.	Insulators		5	5:1
24.	Screw Drivers		5	5:1
25.	Resistors		5	5:1
26.	Capacitors		5	5:1
27.	Logic gates		5	5:1
28.	Inductors		5	5:1
29.	Transistors		5	5:1
30.	Transformers batteries, power supplies		5	5:1
31.	Conducting wires		5	5:1

COMPUTER SOFTWARE

ISCED UNIT CODE: 0619 451 06A

TVET CDACC UNIT CODE: IT/CU/ICTA/CR/01/5/MA

Duration of Unit: 160 hours

Relationship to Occupational Standards

This unit addresses the unit of competency: Install Computer Software

Unit Description:

This unit covers the competencies required to install computer software. It involves the ability to: install computer software, test computer software functionality and perform software maintenance.

Summary of Learning Outcomes

LEARNING OUTCOMES	DURATION (HOURS)
1. Install computer software	70
2. Test computer software functionality	40
3. Perform computer software maintenance	50
TOTAL:	**160**

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Install computer Software	1.1 Identification of computer software 1.1.1 Introduction to computer software 1.1.1.2 Definition of computer software 1.1.1.3 Classification of computer software	• Practical assessment • Project • Observation

	1.1.1.4 Types of computer software 1.1.2 Collecting computer software user needs. 1.2 Selection of computer software 1.2.1 Factors to consider in computer software selection 1.2.2 Acquisition methods of computer software 1.3 Manage local user accounts 1.3.1 Introduction to local user accounts 1.3.2 Types of local user accounts 1.3.2.1 Standard user account 1.3.2.2 Administrator account 1.3.2.3 Guest account 1.3.3 Creating user accounts 1.3.4 Configuration of local user accounts 1.4 Performing data backup 1.4.1 Importance of computer software backup 1.4.2 Types of computer software backup 1.4.3 Back up creation 1.5 Installation of computer Software 1.5.1 Computer software installation media 1.5.2 Computer software installation methods 1.5.3 Types of software registration 1.5.4 Installing computer software 1.5.5 Anti-malware software installation 1.5.5.1 Identify Antimalware to install 1.5.5.2 Identify Antimalware acquisition method 1.5.5.3 Install Antimalware	Checklist • Product Checklist • Written assessment • Portfolio of evidence

	1.5.5.4 Configure Antimalware 1.6 Computer software configuration 1.6.1 Importance of software configuration 1.6.2 Computer software configuration tools	
2. Test computer software functionality.	2.1 Software testing 2.1.1 Importance of software testing 2.1.2 Computer software testing techniques 2.1.3 Computer software testing tools 2.1.3.1 Test Complete 2.1.3.2 Selenium 2.1.3.3 Appium 2.1.3.4 Postman 2.1.4 Performing computer software testing 2.2 Corrective measures 2.2.1 Types of corrective measures 2.2.2 Software corrective tools 2.2.3 Performing corrective measures 2.3 Testing of computer software functionality	• Practical assessment • Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence
3. Perform computer software maintenance.	3.1 Development of Software maintenance schedule 3.1.1 Introduction to computer software maintenance 3.1.1.1 Importance software maintenance 3.1.2 Prepare software maintenance schedule 3.1.3 Types of software maintenance 3.1.3.1 Adaptive 3.1.3.2 Perfective 3.1.3.3 Preventive 3.1.3.4 Corrective 3.1.4 Computer software updates 3.1.4.1 Service packs	• Practical assessment • Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence

	3.1.4.2 Version upgrades 3.1.4.3 Security upgrades 3.1.4.4 Device drivers 3.1.4.5 Utility program updates 3.2 Software functionality monitoring 3.2.1 Software functionality monitoring tools 3.2.2 Operating System event logs 3.2.2.1 Types of event logs 3.2.2.1.1 Error event logs 3.2.2.1.2 Warning event logs 3.2.2.1.3 Information event logs 3.2.2.1.4 Success Audit event logs 3.2.2.1.5 Failure Audit event logs 3.3 Conducting software upgrade 3.3.1 Importance of software upgrade 3.3.2 Types of software upgrade 3.3.3 Conducting software upgrade 3.4 Conducting software update 3.4.1 Importance of software update 3.4.2 Types of software update 3.4.3 Conducting software update 3.5 Observing Safety procedures 3.5.1 Safety measures and procedures 3.5.1.1 Overall/apron/dust coat 3.5.1.2 Antiglare screens 3.5.1.3 Gloves 3.5.2 Personal Protective Equipment's 3.5.2.1 Proper use of tools and equipment	

Suggested Delivery Methods

- Instructor led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainee
- Viewing of related videos
- Group discussions
- Direct instructions

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Trainee: Item)
A	Learning Materials			
21.	Textbooks		5 pcs	5:1
22.	Installation manuals		5 pcs	5:1
23.	Flip Charts		5 pcs	5:1
24.	PowerPoint presentations	For trainer's use		
25.	Installation CDs/DVDs		25pcs	1:1
B	Learning Facilities & infrastructure			
26.	Lecture/theory room		1	25:1
27.	Computer Laboratory		1	25:1
C	Consumable materials			
28.	Printing papers		1 ream	1:20
29.	Foolscaps		1 ream	1:20

30.	Toners		2 pcs	13:1
31.	Assorted colour of whiteboard markers			
D	Tools and Equipment			
32.	Computers		25 pcs	1:1
33.	Projector		1 pcs	25:1
34.	Printers		2 pcs	13:1
35.	Whiteboard		1 pcs	25:1
36.	Flash drives		5 pcs	5:1
37.	External Hard drive		5 pcs	5:1
38.	System Software suite		5 pcs	5:1
39.	Application Software suite		5 pcs	5:1

WORK ETHICS AND PRACTICES

ISCED UNIT CODE: 0417 441 02A

TVET CDACC UNIT CODE: IT/CU/ICTA/BC/01/5/MA

Duration of Unit: 40 hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply work ethics and practices.

Unit Description

This unit covers competencies required to effectively apply work ethics and practices. It involves applying self-management skills, promoting ethical work practices and values, promoting teamwork, maintaining professional and personal development, applying problem-solving and promoting customer care.

Summary of Learning Outcomes

LEARNING OUTCOMES	DURATION (HOURS)
1. Apply self-management skills	**10**
2. Promote ethical practices and values	**10**
3. Promote teamwork	**5**
4. Maintain professional and personal development	**5**
5. Apply problem-solving skills	**5**
6. Promote customer care.	**5**
TOTAL:	**40**

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods

Learning Outcome	**Content**	**Suggested Assessment Methods**
1. Apply self-management skills	1.1 Self-awareness 1.2 Formulating personal vision, mission, and goals 1.3 Healthy lifestyle practices 1.4 Strategies for overcoming work challenge 1.5 Emotional intelligence 1.6 Coping with Work Stress. 1.7 Assertiveness versus aggressiveness and passiveness 1.8 Developing and maintaining high self-esteem 1.9 Developing and maintaining positive self-image 1.10 Time management 1.11 Setting performance targets 1.12 Monitoring and evaluating performance targets	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
2. Promote ethical work practices and values	2.1 Integrity 2.2 Core Values, ethics and beliefs 2.3 Patriotism 2.4 Professionalism 2.5 Organizational codes of conduct 2.6 Industry policies and procedures	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
3. Promote Teamwork	3.1 Types of teams 3.2 Team building 3.3 Individual responsibilities in a	• Observation • Written assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	team 3.4 Determination of team roles and objectives 3.5 Team parameters and relationships 3.6 Benefits of teamwork 3.7 Qualities of a team player 3.8 Leading a team 3.9 Team performance and evaluation 3.10 Conflicts and conflict resolution 3.11 Gender and diversity mainstreaming 3.12 Developing Healthy workplace relationships 3.13 Adaptability and flexibility 3.14 Coaching and mentoring skills	• Third party reports • Portfolio of evidence • Project • Practical
4. Maintain professional and personal development	4.1 Personal vs professional development and growth 4.2 Avenues for professional growth 4.3 Recognizing career advancement 4.4 Training and career opportunities 4.5 Assessing training needs	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	**Content**	**Suggested Assessment Methods**
	4.6 Mobilizing training resources 4.7 Licenses and certifications for professional growth and development 4.8 Pursuing personal and organizational goals 4.9 Managing work priorities and commitments 4.10 Dynamism and on-the-job learning	
5. Apply Problem-solving skills	5.1 Causes of problems 5.2 Methods of solving problems 5.3 Problem-solving process 5.4 Decision making 5.5 Creative thinking and critical thinking process in development of innovative and practical solutions	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
6. Promote Customer Care	6.1 Identifying customer needs 6.2 Qualities of good customer service 6.3 Customer feedback methods 6.4 Resolving customer concerns 6.5 Customer outreach programs 6.6 Customer retention	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Suggested Methods of Instruction

- Instructor lead facilitation of theory using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion
- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Trainee: Item)
A	Learning Materials			
1.	Textbooks		5 pcs	5:1
2.	PowerPoint presentations	For trainer's use		
3.	Assorted colour of whiteboard markers	For trainer's use	2 packets	
4.	e-Didactics	For trainer's use		
5.	Flashcards			
6.	Flip charts			
7.	Whiteboard			
B	Learning Facilities & infrastructure			
8.	Lecture/theory room		1	25:1
C	Consumable materials			

9.	Printing Papers		1 ream	1:20
10.	Toners		2 pcs	13:1
11.	Internet connection			
D	Tools and Equipment			
12.	Projectors		1	25:1
13.	Printers		4	6:1
14.	Computers/Mobile Phones		25 pcs	1:1

NETWORK DESIGN AND MANAGEMENT

ISCED UNIT CODE: 0612 451 07A

TVET CDACC UNIT CODE: IT/CU/ICTA/CR/02/5/MA

Duration of Unit: 200 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform Computer Networking

Unit Description

This unit covers the competencies required to perform network design and management**.** It involves designing computer network, installing computer network, testing computer network and performing computer network maintenance.

Summary of Learning Outcomes

LEARNING OUTCOMES	DURATION (HOURS)
1. Design computer network	**40**
2. Install computer network	**60**
3. Test computer network	**30**
4. Perform computer network maintenance.	**30**
TOTAL	**160**

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Design computer network	1.1 User needs collections 1.1.1 Introduction to computer networking 1.1.1.1 Definition of Computer Network	• Practical assessment • Project

	terms 1.1.2 Computer Network types 1.1.2.1 LAN 1.1.2.2 WAN 1.1.2.3 PAN 1.1.2.4 MAN 1.1.3 Network topologies 1.1.3.1 Star 1.1.3.2 Ring 1.1.3.3 Mesh 1.1.3.4 Hybrid 1.1.3.5 Point to Point 1.1.4 Components of a computer network 1.1.4.1 switches/hubs 1.1.4.2 routers 1.1.4.3 ports 1.1.4.4 computers 1.1.4.5 Transmission media 1.1.5 Computer Network user requirements/needs 1.1.5.1 User requirements identification 1.1.5.2 User requirements analysis 1.1.5.3 User requirements documentation 1.2 Physical network design development 1.3 Logical network design development 1.4 Computer network design 1.4.1 Network design overview 1.4.2 Network design methodology 1.4.2.1 Hierarchical Network Design 1.4.2.2 Flat network 1.4.3 Types of computer network sites (Green field and brownfield) 1.4.4 Network site preparation 1.4.4.1 Network floor plan design	• Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence

	1.4.4.2 Data and Access point 1.4.5 Implement the documented user requirements/needs 1.4.6 Fundamental Design Goals 1.4.6.1 Scalability 1.4.6.2 Availability 1.4.6.3 Security 1.4.6.4 Manageability	
2. Install computer network	2.1 Safety measures 2.1.1 Personal Protective Equipment (PPEs) 2.1.1.1 Overall/apron/dust coat 2.1.1.2 Antiglare screens 2.1.1.3 Dust mask 2.1.1.4 Gloves 2.1.1.5 Antistatic equipment 2.1.1.6 Ergonomics 2.1.1.7 First AID kit 2.1.2 Cable management 2.1.1.8 Proper routing 2.1.1.9 Labelling 2.1.3 Electrical safety 2.1.1.10Use of insulated tools 2.1.1.11Electrical equipment power ratings 2.1.4 Fire safety 2.1.1.12Classes of fires 2.1.1.13Fire extinguishers 2.1.4 Emergency procedures 2.1.1.14First AID kit 2.1.1.15Emergency contact 2.1.1.16Contingency measures 2.2 Computer network components identification 2.2.1 Considerations of network components identification 2.2.1.1 Switches/routers	● Practical assessment ● Project ● Observation Checklist ● Product Checklist ● Written assessment ● Portfolio of evidence

	2.2.1.2 Transmission media and connectors 2.2.1.3 Access points and wireless technology 2.2.1.4 Networking software and management tools 2.2.1.5 Network security devices 2.2.1.6 Servers and storage 2.2.2 Network Tools and materials assembly 2.2.2.1 Basic network tools 2.2.2.1.1 Cable crimpers 2.2.2.1.2 Cable strippers 2.2.2.1.3 Cutters, Scissors, screw drivers Pliers. 2.2.2.1.4 Cable Tie Tools. 2.2.2.1.5 Fiber Optic Tools. 2.2.2.1.6 Insertion - Extraction Tools. 2.2.2.1.7 Manual/Automatic Switch Boxes. 2.2.2.1.8 Network Testers. 2.2.2.1.9 Punch down Tools. 2.2.2.1.10 Tools usage and safety 2.2.2.1.11 Driver installers 2.2.2.1.12 Multimeter 2.2.2.1.13 Tone generator and probe 2.2.3 Computer Network materials 2.2.3.1 Network cables 2.2.3.2 Cable trunking covers 2.2.3.3 Connectors 2.2.3.4 RJ45 Sockets 2.2.3.5 Patch cords 2.2.3.6 Cable ties 2.3 Computer network set up	

	2.3.1 Network cabling and installation 2.3.1.1 Network design layout 2.3.1.2 Understanding cabling standards and codes 2.3.1.3 Cable termination and installation 2.3.1.4 Setting up wireless network devices 2.3.1.5 Network set up as per the design 2.3.1.6 Application of cable management best practices 2.4 Computer network devices configuration 2.4.1 Network models (TCP/IP, OSI) 2.4.2 Understanding IP Addressing 2.4.2.1 Classful IP Addressing 2.4.2.2 TCP/IP addressing 2.4.2.3 IPV4 and IPV6 2.4.2.4 IP Address Classes 2.4.2.5 Classless interdomain routing (CIDR-Subnetting) 2.4.2.6 Select IP addressing scheme (static vs. dynamic). 2.4.3 Basic switch and router configuration 2.4.3.1 Initial set up and configuration 2.4.3.2 Configuring interfaces and IP addresses 2.4.3.3 Setting up routing protocols (EIGRP, RIP and OSPF) 2.4.3.4 Configuring VLANs 2.4.3.5 Configuring access control lists 2.4.3.6 Implementing network address translation (NAT) and port address translation (PAT) 2.4.3.7 Implementing port security 2.4.3.8 Implementing spanning tree protocol (STP).	

	2.4.3.9 Configuration link aggregation (LACP) 2.4.4 Wireless access point configuration 2.4.4.1 Setting up access points (APs) 2.4.4.2 SSID, DHCP, DNS, SMTP 2.4.4.3 Configuring wireless security 2.4.4.4 Managing wireless network 2.4.4.5 Network Security configuration 2.4.4.6 Definition of Network privileges 2.4.4.7 Implement firewall and security policies 2.4.4.8 Types of Privileged Accounts 2.4.4.9 Network privileges are allocated according to the network configuration. 2.5 Computer network documentation 2.5.1 Define network documentation 2.5.2 Importance of network documentation 2.5.3 Types of network documentations 2.5.3.1 Physical, Logical and configuration 2.6 Computer network components disposal 2.6.1 Identify computer network waste 2.6.2 Classify computer network waste 2.6.2.1 E- waste 2.6.2.2 Hazards 2.6.2.3 Disposal methods 2.6.3 Legal regulation and compliance on waste disposal 2.6.3.1 Waste management act, 2022 2.6.3.2 EMCA act, 2015 on waste management 2.6.4 Disposal methods 2.6.4.1 The public procurement and assets disposal act, 2015	

3. Test computer network	3.1 Introduction to network testing 3.1.1 Importance of network testing 3.1.2 Network testing tools and equipment 2.6.4.2 Clamp meter 2.6.4.3 Voltmeter 2.6.4.4 Cable tester 2.6.4.5 Signal tester 2.6.4.6 Ping 2.6.4.7 Traceroute 2.6.4.8 Wireshark 3.2 Network components testing 3.2.1 Types of network testing 2.6.4.9 Performance 2.6.4.10Functional 2.6.4.11Security 3.2.2 Network testing procedures and standards 3.3 Network testing report 3.3.1 Importance of generating network test report 3.3.2 Components of a network test report 3.3.3 Presenting network test reports 2.6.4.12Reports presentation techniques 2.6.4.13Preparing interactive presentations	• Practical assessment • Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence
4. Perform computer network maintenance.	4.1 Computer network maintenance schedule 4.1.1 Importance of network maintenance 4.1.2 Preparation of maintenance schedule 4.1.3 Network troubleshooting process 4.1.4 Network troubleshooting techniques 4.2 Computer network Monitoring 4.2.1 Monitoring tools 4.2.1.1 Ping 4.2.1.2 Tracert	• Practical assessment • Project • Observation Checklist • Product Checklist • Written

	4.2.1.3 NSLookup 4.2.1.4 Ipconfig 4.2.1.5 Speed test 4.2.1.6 Traceroute 4.2.1.7 Wireshark 4.2.2 Setting and configuring monitoring tools 4.2.3 Analysing network performance data 4.3 Computer network optimization 4.3.1 Network optimization techniques 4.3.2 Implementing quality of service (QOS) 4.4 Computer network maintenance report 4.4.1 Importance of generating network maintenance report 4.4.2 Components of a network maintenance report 4.4.3 Preparation of network maintenance report	assessment • Portfolio of evidence

Suggested Delivery Methods

- Instructor led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainee
- Viewing of related videos
- Group discussions
- Direct instructions

Recommended Resources for 25Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Trainee: Item)
A	Learning Materials			
11.	Textbooks		5 pcs	5:1

12.	Installation manuals			
13.	Charts			
14.	PowerPoint presentations	For trainer's use		
B	Learning Facilities & infrastructure			
15.	Lecture/theory room		1	25:1
16.	Computer laboratory		1	25:1
C	Consumable materials			
17.	5Printing papers		1 ream	1:20
18.	Toners		2 pcs	13:1
19.	Assorted colour of whiteboard markers			
D	Tools and Equipment			
14.	Computers		25 pcs	1:1
15.	Projector		1 pc	25:1
16.	Signal testers		5 pcs	5:1
17.	Header checker		25 pcs	1:1
18.	Crimping tools		25 pcs	1:1
19.	Cable tester		5 pcs	5:1
20.	Punch Downs		5 pcs	5:1
21.	Switches		5pcs	5:1
22.	Repeaters		5pcs	5:1
23.	Routers/modem		5pcs	5:1
24.	Network tool kit		25 pcs	1:1
25.	Gateways		5pcs	5:1
26.	Packets of RJ45		300 pcs	1:10
27.	Fibre Modules (SFP)		5pcs	5:1
28.	UTP Ethernet Cable		300 metres	1:10
29.	25 Antistatic gloves		25 pairs	1:1

MODULE 4

UNIT CATEGORY	ISCED UNIT CODE	TVET CDACC UNIT CODE	UNIT NAME	DURATION (HOURS)
COMMON	0613 441 05A	IT/CU/ICTA/CC/02/5/MA	Computer Programming Principles	180
CORE	0612 451 08A	IT/CU/ICTA/CR/01/5/MA	Computerized Database System	200
BASIC	0031 441 01A	IT/CU/ICTA/BC/01/5/MA	Communication Skills	40
BASIC	0413 441 03A	IT/CU/ICTA/BC/02/5/MA	Entrepreneurial Skills	40
	Sub-Total Hours			**460**
	Industrial Training			**480**

COMPUTER PROGRAMMING PRINCIPLES

ISCED UNIT CODE: 0613 451 05A

TVET CDACC UNIT CODE: IT/CC/ICTA/CC/02/5/MA

Duration of Unit: 180 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply Computer Programming Principles

Unit Description

This unit covers the competencies required to apply computer programming principles. It involves applying computer programming skills, demonstrating structured programming skills and demonstrating object-oriented programming skills.

Summary of Learning Outcomes

LEARNING OUTCOMES	DURATION (HOURS)
1. Apply Computer programming skills	50
2. Demonstrate Structured programming skills	60
3. Demonstrate Object-oriented programming skills	70
TOTAL	**180**

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	**Content**	**Suggested Assessment Methods**
1. Apply computer programming skills	1.1 Identification of Programming Languages 1.1.1 Overview of programming language categories (e.g., procedural, object-oriented, functional) 1.1.2 Criteria for selecting languages based on	• Practical Activities • Project work • Demonstration • Group discussions • Observation

	user requirements 1.2 Application Programming Paradigms 1.2.1.1 Explanation of common programming paradigms 1.2.1.2 Functional 1.2.1.3 Procedural 1.2.1.4 Object-oriented 1.2.1.5 Imperative 1.2.1.6 Declarative 1.2.2 Choosing the appropriate paradigm based on project needs 1.3 Program Development Life Cycle 1.3.1 Stages of the program development life cycle 1.3.2 Best practices for adapting the life cycle to work requirements 1.4 Application of Program Design Tools 1.4.1 Overview of design tools 1.4.1.1 Flow charts 1.4.1.2 Decision tables 1.4.1.3 Decision trees 1.4.1.4 Pseudocode 1.4.1.5 Algorithm 1.4.2 Selecting design tools based on user requirements and project complexity 1.5 Identification of Program Writing Tools 1.5.1 Common program writing tools and IDEs 1.5.1.1 Text editors 1.5.1.2 Compilers Linkers 1.5.1.3 Debuggers 1.5.1.4 Special Integrated Development	• Portfolio of Evidence • Written tests

	Environment (IDE) 1.5.1 Evaluating tools based on system requirements and developer preferences	
2. Demonstrate structured programming skills	2.1 Declaration of Identifiers in C language 2.1.1 Guidelines for naming conventions and best practices 2.1.2 Ensuring identifiers align with program design specifications 2.2 Initializing Variables and Constants in C language 2.2.1 Importance of proper initialization in programming 2.2.2 Techniques for initialization based on design specifications 2.3 Applying User-Defined Data Types in C language 2.3.1 Overview of user-defined data types in C language 2.3.1.1 Structures 2.3.1.2 Classes 2.3.1.3 Arrays 2.3.1.4 Function 2.3.2 Criteria for selecting data types based on system requirements 2.4 Creating Computer program input in C language 2.5 Application of Data control structures in C program 2.5.1 Types of control structures 2.5.1.1 Selection 2.5.1.2 Loops	• Practical Activities • Project work • Demonstration • Group discussions • Observation • Third Party report • Portfolio of Evidence • Written tests

	2.5.1.3 Sequence 2.5.2 Best practices for implementing control structures as per design requirements 2.6 Data structures in C program 2.6.1 Overview of common data structures. 2.6.1.1 Arrays 2.6.1.2 Queue 2.6.1.3 Stack 2.6.1.4 Linked lists 2.6.2 Selecting appropriate data structures based on design specifications. 2.7 Creating C computer program subroutines 2.7.1 Benefits of using subroutines 2.7.2 Designing subroutines to meet user needs 2.7.3 Functions and subprograms 2.8 Coding of C Computer program output 2.9 Performing C Computer Program Debugging 2.9.1 Common debugging techniques and tools 2.9.2 Following work procedures for systematic debugging 2.10 Compiling C Computer Program 2.10.1 Steps involved in the compilation process 2.10.2 Ensuring compliance with system requirements during compilation	

3. Demonstrate object-oriented programming skills	3.1 Implementing Objects and Classes in C++ language 3.1.1 Overview of objects and classes in OOP 3.1.2 Ensuring implementation aligns with work procedures 3.2 Declaring Object Methods in C++ language 3.2.1 Defining methods that fulfill application requirements 3.2.2 Best practices for method naming and functionality 3.3 Applying Namespaces in C++ language 3.3.1 Understanding the role of namespaces in OOP 3.3.2 Implementing namespaces 3.4 Data abstraction concepts in C++ language 3.4.1 Definition of data abstraction 3.4.2 Importance of data abstraction 3.4.3 Implementing of data abstraction in OOP 3.5 Object encapsulations in C++ language 3.5.1 Definition of Object encapsulations 3.5.2 Importance of Object encapsulations 3.5.3 Implementing of Object encapsulations in OOP 3.6 Class templates implementation 3.7 Class inheritance implementation 3.7.1 Definition of data abstraction 3.7.2 Importance of data abstraction 3.7.3 Base class 3.7.4 Derived class	• Practical Activities • Project work • Demonstration • Group discussions • Observation • Third Party report • Portfolio of Evidence • Written tests

	3.7.5 Inheritance relationships 3.7.6 Types of inheritance 3.8 Implementing class polymorphism in C++ language 3.8.1 Definition of data polymorphism 3.8.2 Importance of data polymorphism 3.8.3 Implementing of data polymorphism in OOP	

Suggested Delivery Methods

- Instructor led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainee
- Viewing of related videos
- Group discussions
- Direct instructions

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Trainee: Item)
A	Learning Materials			
1.	Textbooks		5 pcs	1:5
2.	Installation manuals			
3.	Charts			
4.	PowerPoint presentations	For trainer's use		
5.	Assorted colour of whiteboard markers	For trainer's use		

6.	e-Didactics	For trainer's use		
B	Learning Facilities & infrastructure			
7.	Lecture/theory room		1	1:25
8.	Computer Laboratory		1	1:25
C	Consumable materials			
9.	Printing Papers		1 ream	1:20
10.	Toners		2 pcs	13: 1
11.	Internet connection			
D	Tools and Equipment			
12.	Projectors		1	25:1
13.	Printers		4	6:1
14.	Flash drives		5 pcs	5:1
15.	Computers		25 pcs	1:1
16.	Integrated Development Environment (IDEs) – C,C++, Java and Visual Studio, IntelliJ IDEA, Python IDE		25 pcs	1:1

COMPUTERIZED DATABASE SYSTEMS

ISCED UNIT CODE: 0612 451 08A

TVET CDACC UNIT CODE: IT/CU/ICTA/CR/03/5/MA

Duration of Unit: 200 Hours

Relationship to Occupational Standards

This unit addresses the unit of competency: Manage Computerized Database Systems

Relationship to Occupational Standards

This unit addresses the unit of competency: Manage Computerized Database Systems

Unit Description:

This unit covers the competencies required to manage computerized database systems. It involves designing computerized database, creating computerized database, manipulating computerized database, testing computerized database and maintaining computerized database.

Summary of Learning Outcomes:

Learning Outcomes	Durations (Hours)
1. Design computerized database.	30
2. Create a computerized database.	60
3. Manipulate computerized database.	60
4. Test computerized database.	30
5. Maintain computerized database.	20
Total Hours	**200**

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment

		Method
1. Design computerized database.	1.1 Establishing computerized database requirements 1.1.1 Introduction to DBMS 1.1.1.1 Definition of terms 1.1.1.2 Functions of database systems 1.1.1.3 Database applications 1.1.1.4 Types of database management system 1.1.1.5 Major components of the DBMS environment. 1.1.1.6 Database architecture 1.1.2 Database requirements gathering techniques 1.1.2.1 Database requirement elicitation 1.1.2.2 Database assessment requirement report 1.2 Database models 1.2.1 Relational Models 1.2.2 Hierarchical models 1.2.3 Network Models 1.2.4 Object-oriented models 1.2.5 Dimensional model 1.3 Database design 1.3.1 Conceptual design	• Practical assessment • Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence

	1.3.2 ERDs 1.3.2.1 Types of entities and notations 1.3.2.2 Types of attributes and notations 1.3.2.3 Types of relationship and notation 1.3.3 Relational model 1.3.4 Logical design 1.3.4.1 Define tables 1.3.4.2 Keys 1.3.4.3 Normalization 1.3.5 Physical design 1.3.5.1 Selection of data types 1.3.5.2 Indexing 1.3.5.3 constrains	
2. Create computerized database.	2.1 Creation of database objects 2.1.1 Database objects 2.1.1.1 Tables 2.1.1.2 Forms 2.1.1.3 Queries 2.1.1.4 Reports 2.1.1.5 Macros 2.1.2 Fields 2.1.2.1 Records 2.1.2.2 Data types	• Practical assessment • Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence

	2.1.2.3 Field properties 2.1.2.4 Keys 2.2 Database development 2.2.1 Creation of tables 2.2.2 Linking of tables 2.2.3 Creation of Queries 2.2.4 Creation of forms 2.2.5 Creation of reports 2.2.6 Integrity constraints 2.2.6.1 Nulls 2.2.6.2 Entity Integrity 2.2.6.3 Referential Integrity 2.2.6.4 General Constraints 2.2.7 Database indexing 2.2.7.1 Primary indexing 2.2.7.2 Secondary indexing 2.3 Data relationships 2.3.1.1 A one-to-one relationship 2.3.1.2 one-to-many relationship 2.3.1.3 Many-to-one relationship 2.3.1.4 A many-to-many	

	relationship	
3. Manipulate computerized database.	3.1 Database population 3.1.1 Database operations 2.3.1.5 Insert 2.3.1.6 Select 2.3.1.7 Update 2.3.1.8 Delete 3.2 Data retrieval 3.2.1 Database Querying Languages – SQL 2.3.1.9 Introduction to SQL 2.3.1.10 Writing SQL commands 2.3.1.11 SQL Views 3.3 Database report Generation 3.3.1 Generate report 3.3.2 Edit report 3.3.3 Format report 3.3.4 Apply header and footer 3.3.5 Print report	• Practical assessment • Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence
4. Test computerized database.	4.1 Test plan development 4.1.1 Importance of database testing 4.1.2 Types of database testing 4.1.2.1 Structural DB testing 4.1.2.1.1 Schema 4.1.2.1.2 Table and	• Practical assessment • Project • Observation Checklist • Product Checklist • Written

	column 4.1.2.2 Functional testing 4.1.2.2.1 Whitebox 4.1.2.2.2 Blackbox 4.1.2.3 Stored procedure 4.1.2.4 Non-functional testing 4.1.2.4.1 Load testing 4.1.2.4.2 Stress testing 4.2 Prepare test data 4.3 Perform database testing using test data 4.3.1 Validate database results 4.4 Generate test report 4.4.1 Importance of test reports 4.4.2 Prepare test report	assessment • Portfolio of evidence
5. Maintain computerized database.	5.1 Database optimization 5.1.1 Introduction to database optimization 5.1.2 Query optimization techniques 5.1.3 Indexing strategies 5.1.4 Partitioning and storage management 5.2 Computerized database maintenance schedule preparation 5.2.1 Types of Database maintenance techniques 5.2.2 Database maintenance schedule preparation	• Practical assessment • Project • Observation Checklist • Product Checklist • Written assessment • Portfolio of evidence

	5.2.3 Key elements in preparation of maintenance schedule 5.2.4 Computerized database backup and recovery 5.3 Computerized database monitoring 5.3.1 Computerized database monitoring tools 5.3.2 Resource Utilization 5.3.3 Query Performance 5.3.4 Transaction log monitoring 5.3.5 Connection monitoring 5.3.6 Alerting systems 5.4 Generation of Computer database maintenance report	

Suggested Delivery Methods

- Demonstration by trainer
- Practical work by trainee
- Viewing of related videos
- Group discussions
- Direct instructions
- Instructor led facilitation using active learning strategies

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Trainee: Item)

A	**Learning Materials**			
1.	Textbooks	For trainee's use	5pcs	5:1
2.	Installation manuals	For trainer's use		
3.	Charts	For trainer's use		
4.	PowerPoint presentations	For trainer's use		
B	**Learning Facilities & infrastructure**			
1.	Lecture/theory room	For training	1	25:1
2.	Computer Laboratory	For training	1	25:1
C	**Consumable materials**			
3.	Printing papers	For printing	1 ream	1:20
4.	Toners	For printers	2 pcs	13:1
5.	Assorted colour of whiteboard markers	For trainer's use		
D	**Tools and Equipment**			
6.	Computers	For training	25 pcs	1:1
7.	Projector	For trainer's use	1pc	25:1
8.	Printers	For printing	5 pcs	5:1
9.	Whiteboard	For trainer's use	1pc	25:1
10.	flash drives	For training	5 pcs	5:1
11.	External Hard drive	For training	5 pcs	5:1
12.	Microsoft Access	For training	25 pcs	1:1
13.	MYSQL	For training	25 pcs	1:1

14.	Test Data Generator	For training	25 pcs	1:1
15.	WAMP/XAMP	For training	25 pcs	1:1

COMMUNICATION SKILLS

ISCED UNIT CODE: 0031 441 01A

TVET CDACC UNIT CODE: IT/CU/ICTA/BC/01/5/MA

Duration of Unit: 40 hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply Communication Skills

Unit Description

This unit covers the competencies required to apply communication skills. It involves applying communication channels, written, non-verbal, oral, and group communication skills.

Summary of Learning Outcomes

LEARNING OUTCOMES	DURATION (HOURS)
1. Apply communication channels.	5
2. Apply written communication skills.	10
3. Apply non-verbal skills.	10
4. Apply oral communication skills.	5
5. Apply group communication skills.	10
TOTAL	**40**

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply communication channels	1.1 Communication process 1.1.1 Principles of effective communication 1.2 Channels/medium/modes of communication	• Oral questions • Written assessment • Observation • Portfolio of Evidence • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	1.1.1 Factors to consider when selecting a channel of communication 1.1.2 Barriers to effective communication 1.2 Flow/patterns of communication 1.2.1 Sources of information 1.2.2 Organizational policies	● Third party report
2. Apply written communication skills	2.1 Types of written communication 2.2 Elements of communication 2.3 Organization requirements for written communication	● Oral assessment ● Written assessment ● Observation ● Portfolio of Evidence ● Practical assessment ● Third party report
3. Apply non-verbal communication skills	3.1 Utilize body language and gestures 3.2 Apply body posture 3.3 Apply workplace dressing code	● Oral assessment ● Written assessment ● Observation ● Portfolio of Evidence ● Practical assessment ● Third party report
4. Apply oral communication skills	4.1 Types of oral communication pathways 4.2 Effective questioning techniques 4.3 Workplace etiquette	● Oral assessment ● Written assessment ● Observation ● Portfolio of Evidence ● Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	4.4 Active listening	● Third party report
5. Apply group discussion skills	5.1 Establishing rapport 5.2 Facilitating resolution of issues 5.3 Developing action plans 5.4 Group organization techniques 5.5 Turn-taking techniques 5.6 Conflict resolution techniques 5.7 Team-work	● Oral assessment ● Written assessment ● Observation ● Portfolio of Evidence ● Practical assessment

Suggested Methods of Instruction

- Roleplaying
- Simulation
- Field trips
- Viewing of related videos
- Demonstrations
- Online Training
- Group discussions.
- Instructor led facilitation using active learning strategies.

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Trainee: Item)
A	Learning Materials			
1.	Textbooks		5 pcs	5:1
2.	PowerPoint presentations	For trainer’s use		

3.	Assorted colour of whiteboard markers	For trainer's use	2 packets	
4.	e-Didactics	For trainer's use		
5.	Flashcards			
6.	Flip charts			
7.	Whiteboard			
B	Learning Facilities & infrastructure			
8.	Lecture/theory room		1	25:1
C	Consumable materials			
9.	Printing Papers		1 ream	1:20
10.	Toners		2 pcs	13:1
11.	Internet			
D	Tools and Equipment			
12.	Projectors		1	25:1
13.	Printers		4	6:1
14.	Computers/Smartphones		25 pcs	1:1

ENTREPRENEURIAL SKILLS

ISCED UNIT CODE: 0413 441 03A

TVET CDACC UNIT CODE: IT/CU/ICTA/BC/02/5/MA

Duration of unit: 70 hours

Relationship to occupational standards

This unit addresses the unit of competency: Apply Entrepreneurial skills.

Unit Description:

This unit covers the competencies required to demonstrate an understanding of entrepreneurship. It involves the ability to: apply financial literacy, apply entrepreneurial concepts, identify entrepreneurship opportunities, apply business legal aspects, innovate business strategies, and develop business plans.

Summary of Learning Outcomes

LEARNING OUTCOMES	DURATION (HOURS)
1. Apply financial literacy	5
2. Apply the entrepreneurial concept	5
3. Identify entrepreneurship opportunities	5
4. Apply business legal aspects	10
5. Innovate Business Strategies	5
6. Develop business plan	10
TOTAL	**40**

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply financial literacy	1.1 Personal finance management	• Observation • Project

Learning Outcome	Content	Suggested Assessment Methods
	1.2 Balancing between needs and wants 1.3 Budget Preparation 1.4 Savings management 1.5 Factors to consider when deciding where to save 1.6 Debt management 1.7 Factors to consider before taking a loan 1.8 Investment decisions 1.9 Types of investments 1.10 Factors to consider when investing money 1.11 Insurance services • Insurance products available in the market • Insurable risks	• Written assessment • Oral assessment • Third party report • Interviews
2. Apply entrepreneurial concept	2.1 Difference between Entrepreneurs and Business persons 2.2 Types of entrepreneurs 2.3 Ways of becoming an entrepreneur 2.4 Characteristics of Entrepreneurs 2.5 salaried employment and self-	• Observation • Project • Written assessment • Oral assessment • Third party report

Learning Outcome	Content	Suggested Assessment Methods
	employment 2.6 Requirements for entry into self-employment 2.7 Roles of an Entrepreneur in an enterprise 2.8 Contributions of Entrepreneurship	
3. Identify entrepreneurship opportunities	3.1 Sources of business ideas 3.2 Factors to consider when evaluating business opportunity 3.3 Business life cycle	• Observation • Project • Written assessment • Oral assessment • Third party report
4. Apply business legal aspects	4.1 Forms of business ownership 4.2 Business registration and licensing processing 4.3 Types of contracts and agreements 4.4 Employment laws 4.5 Taxation laws	• Observation • Project • Written assessment • Oral assessment • Third party report
5. Innovate business Strategies	5.1 Creativity in business 5.2 Innovative business strategies 5.3 Entrepreneurial Linkages 5.4 ICT in business growth and development	• Observation • Project • Written assessment • Oral assessment • Third party report

Learning Outcome	Content	Suggested Assessment Methods
6. Develop Business Plan	6.1 Business description 6.2 Marketing plan 6.3 Organizational/Management plan 6.4 Production/operation plan 6.5 Financial plan 6.6 Executive summary 6.7 Business plan presentation 6.8 Business idea incubation	• Observation • Written assessment • Project • Oral assessment • Third party report

Suggested Methods of Instruction

- Direct instruction with active learning strategies
- Project (Business plan)
- Case studies
- Field trips
- Group Discussions
- Demonstration
- Question and answer
- Problem solving
- Experiential
- Team training
- Guest speakers

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/	Quantity	Recommended

		Specifications		Ratio (Trainee: Item)
A	Learning Materials			
1.	Textbooks		5 pcs	5:1
2.	Business plan templates		5 pcs	5:1
3.	Business Journals		5 pcs	5:1
4.	Newspapers and Handouts			
5.	PowerPoint presentations	For trainer's use		
6.	Assorted colour of whiteboard markers	For trainer's use	2 packets	
7.	e-Didactics	For trainer's use		
8.	Flashcards			
9.	Flip charts			
10.	Whiteboard			
B	Learning Facilities & infrastructure			
11.	Lecture/theory room		1	25:1
C	Consumable materials			
12.	Printing Papers		1 ream	1:20
13.	Toners		2 pcs	13:1
14.	Internet connection			
D	Tools and Equipment			
15.	Projectors		1	25:1

16.	Printers		4	6:1
17.	Computers/Smartphones		25 pcs	1:1